



3.4 Differentiating Inverse Trigonometric Functions

3.5 Selecting Procedures for Calculating Derivatives

3.6 Calculating Higher Order Derivatives

WARM UP!

Ex 1: Evaluate.

a) $\tan^{-1}(\sqrt{3})$	b) $\cos^{-1}\left(\frac{\sqrt{3}}{2}\right)$	c) $\sin^{-1}(-1)$
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Derivatives of Inverse Trigonometric Functions

$\frac{d}{dx}[\sin^{-1} x] =$	$\frac{d}{dx}[\csc^{-1} x] =$
$\frac{d}{dx}[\cos^{-1} x] =$	$\frac{d}{dx}[\sec^{-1} x] =$
$\frac{d}{dx}[\tan^{-1} x] =$	$\frac{d}{dx}[\cot^{-1} x] =$



Do you notice a pattern? The derivatives of inverse trigonometric and their corresponding cofunctions form _____ pairs! Think of these derivatives as BOGO deals.

Ex 1:

If $f(x) = \sin^{-1} x$, then $f'\left(\frac{\sqrt{3}}{2}\right) =$

- (A) $\frac{\pi}{6}$ (B) $\frac{\pi}{3}$ (C) $\frac{4}{7}$ (D) 2

Ex 2: Find the derivative.

a) $y = \sin^{-1}(2x)$	b) $f(x) = \sec^{-1}(e^x)$
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Ex 3: Write the equation of the tangent line to $f(x) = \tan^{-1}\left(\frac{x}{2}\right)$ at $x = -2$.

Ex 4:

If $\lim_{h \rightarrow 0} \frac{\arccos(a+h) - \arccos(a)}{h} = -3$, which of the following could be the value of a ?

(A) $\frac{\sqrt{8}}{3}$

(B) $\frac{1}{3}$

(C) 3

(D) 1

**This problem was authored by the Algebras*

Ex 5:

If $\arctan y = \ln x$, then $\frac{dy}{dx} =$

(A) $\tan\left(\frac{1}{x}\right)$

(B) $\tan(\ln x)$

(C) $\frac{1+y^2}{xy}$

(D) $\frac{x}{1+y^2}$

(E) $\frac{1+y^2}{x}$

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Higher Order Derivatives

Higher order derivatives refer to the repeated process of differentiating a function multiple times.

	Lagrange	Leibniz
Function	$f(x)$	y
1 st Derivative	$f'(x)$	$\frac{dy}{dx}$
2 nd Derivative	$f''(x)$	$\frac{d^2y}{dx^2}$
3 rd Derivative	$f'''(x)$	$\frac{d^3y}{dx^3}$
4 th Derivative	$f^{(4)}(x)$	$\frac{d^4y}{dx^4}$
n th Derivative	$f^{(n)}(x)$	$\frac{d^ny}{dx^n}$



Reminder:
 $f^{(4)}(x) \neq f^4(x)$.
 $f^{(4)}(x)$ is the 4th derivative of f
while $f^4(x)$ is the 4th power of f .

Be Careful!
In most cases finding higher order derivatives requires an iterative process. If the n^{th} derivative is incorrect, the $(n+1)^{\text{th}}$ and all other subsequent derivatives will also be incorrect!



Ex 6: Find the indicated derivative.

a) $f(x) = 3x^4 + 2$, $f'''(x) =$	b) $y = \sin x$, $\frac{d^2y}{dx^2} =$
c) $y = \sin x$, $\frac{d^7y}{dx^7} =$	d) $y = \sin x$, $\frac{d^{2025}y}{dx^{2025}} =$
e) $y = e^x$, $y^{(123456789)} =$	f) $g(x) = \cos x$, $g^{(163)}(x) =$
g) $y = \cos(2x)$, $\frac{d^{163}y}{dx^{163}} =$	h) $h(t) = \csc t$, $h''(t) =$



Ex 7:

If $y = \cos x - \ln(2x)$, then $\frac{d^3y}{dx^3} =$

- (A) $\sin x - \frac{2}{x^3}$
(B) $-\sin x - \frac{2}{x^3}$
(C) $\sin x - \frac{1}{x^3}$
(D) $-\sin x - \frac{1}{x^3}$

Ex 8: Use a graphing utility to find the indicated values.

a) $\frac{d}{dx}(\ln x)|_{x=8}$

b) $f(x) = \arctan x, f'(5)$

c) If $y = \frac{3x+2}{\sqrt{x^2+5}}, \frac{d^2y}{dx^2}|_{x=\frac{1}{2}}$

Strategies for Selecting Procedures for Calculating Derivatives

- Read the problem out loud, reciting all operations.
- Once the operations have been identified, determine what rule or rules are needed to differentiate.
- Decide if its best to rewrite before differentiating.
- Students often forget to use product and or use chain rule incorrectly. Don't be those students! 🙄